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Inventor(s)	Hwang; Sungmin et al.

Semiconductor device and data storage system including the same

Abstract

A semiconductor device and a data storage system including the same, the semiconductor device including a substrate structure; a stack structure; a vertical memory structure; a vertical dummy structure; and an upper separation pattern, wherein hen viewed on a plane at a first height level, higher than a height level of a lowermost end of the upper separation pattern, the dummy channel layer includes a first dummy channel region facing the dummy data storage layer and a second dummy channel region facing the dummy data storage layer, the first dummy channel region having a thickness different from a thickness of the second dummy channel region.

Inventors: **Hwang; Sungmin (Hwaseong-si, KR), Kang; Taewon (Seoul, KR), Woo; Dongsung (Suwon-si, KR), Lee; Taegon (Seoul, KR), Park; Bongtae (Seoul, KR), Shim; Jaejoo (Suwon-si, KR)**

Applicant: **SAMSUNG ELECTRONICS CO., LTD.** (Suwon-si, KR)

Family ID: **84976834**

Assignee: **SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)**

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Primary Examiner: Ojha; Ajay

Assistant Examiner: Chiu; Tsz K

Attorney, Agent or Firm: Muir Patent Law, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This application claims benefit of priority to Korean Patent Application No. 10-2021-0096984 filed on Jul. 23, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.
BACKGROUND
1. Field
(2) Embodiments relate to a semiconductor device and a data storage system including the same.
2. Description of the Related Art
(3) A semiconductor device for storing high-capacity data in an electronic system requiring data storage has been considered. Accordingly, measures for increasing the data storage capacity of a semiconductor device have been considered. For example, as one method for increasing data storage capacity of a semiconductor device, a semiconductor device including memory cells arranged three-dimensionally, rather than memory cells arranged two-dimensionally, has been considered.
SUMMARY
(4) The embodiments may be realized by providing a semiconductor device including a substrate structure; a stack structure on the substrate structure and including interlayer insulating layers and gate electrodes alternately and repeatedly stacked in a vertical direction that is perpendicular to an upper surface of the substrate structure; a vertical memory structure penetrating through the stack structure in the vertical direction; a vertical dummy structure penetrating through the stack structure in the vertical direction; and an upper separation pattern on the stack structure, the upper separation pattern including a first portion and a second portion extending in a first direction, parallel to the upper surface of the substrate structure, the first portion intersecting the vertical dummy structure and the second portion extending from the first portion and penetrating through a portion of the stack structure, wherein the second portion of the upper separation pattern penetrates through a plurality of the gate electrodes, the vertical memory structure includes an insulating region, a channel layer on a side surface of the insulating region, a first dielectric layer on an external side surface of the channel layer, a data storage layer on an external side surface of the first

dielectric layer, a second dielectric layer on an external side surface of the data storage layer, and a pad pattern on the insulating region, the vertical dummy structure includes a dummy insulating region, a dummy channel layer on a side surface of the dummy insulating region, a first dummy dielectric layer on an external side surface of the dummy channel layer, a dummy data storage layer on an external side surface of the first dummy dielectric layer, a second dummy dielectric layer on an external side surface of the dummy data storage layer, and a dummy pad pattern on the dummy insulating region, and when viewed on a plane at a first height level, higher than a height level of a lowermost end of the upper separation pattern, the dummy channel layer includes a first dummy channel region facing the dummy data storage layer and a second dummy channel region facing the dummy data storage layer, the first dummy channel region having a thickness different from a thickness of the second dummy channel region.

(5) The embodiments may be realized by providing a semiconductor device including a substrate structure; a stack structure on the substrate structure and including interlayer insulating layers and gate electrodes alternately and repeatedly stacked in a vertical direction that is perpendicular to an upper surface of the substrate structure; a vertical memory structure penetrating through the stack structure in the vertical direction; a vertical dummy structure penetrating through the stack structure in the vertical direction; an upper separation pattern on the stack structure, the upper separation pattern including a first portion and a second portion extending in a first direction, parallel to the upper surface of the substrate structure, the first portion intersecting the vertical dummy structure and the second portion extending from the first portion and penetrating through a portion of the stack structure; a contact plug in contact with the vertical memory structure and on the vertical memory structure; and a bitline electrically connected to the contact plug and on the contact plug, wherein the second portion of the upper separation pattern penetrates through a plurality of upper gate electrodes of the gate electrodes, the vertical memory structure includes an insulating region, a channel layer on a side surface of the insulating region, a first dielectric layer on an external side surface of the channel layer, a data storage layer on an external side surface of the first dielectric layer, a second dielectric layer on an external side surface of the data storage layer, and a pad pattern on the insulating region, the vertical dummy structure includes a dummy insulating region, a dummy channel layer on a side surface of the dummy insulating region, a first dummy dielectric layer on an external side surface of the dummy channel layer, a dummy data storage layer on an external side surface of the first dummy dielectric layer, a second dummy dielectric layer on an external side surface of the dummy data storage layer, and a dummy pad pattern on the dummy insulating region, and when viewed on a plane at a first height level that higher than a height level of a lowermost surface of a lowermost one of the plurality of upper gate electrodes, and lower than a height level of a lowermost surface of the pad pattern, a thickness of the dummy channel layer of the vertical dummy structure is smaller than a thickness of the channel layer of the vertical memory structure.

(6) The embodiments may be realized by providing a data storage system including a semiconductor device including an input/output pad; and a controller electrically connected to the semiconductor device through the input/output pad and configured to control the semiconductor device, wherein the semiconductor device includes a substrate structure, a stack structure on the substrate structure and including interlayer insulating layers and gate electrodes alternately and repeatedly stacked in a vertical direction, perpendicular to an upper surface of the substrate structure, a vertical memory structure penetrating through the stack structure in the vertical direction, a vertical dummy structure penetrating through the stack structure in the vertical direction, an upper separation pattern on the stack structure, the upper separation pattern including a first portion and a second portion extending in a first direction, parallel to the upper surface of the substrate structure, the first portion intersecting the vertical dummy structure and the second portion extending from the first portion and penetrating through a portion of the stack structure, a contact plug in contact with the vertical memory structure and on the vertical memory structure,

and a bitline electrically connected to the contact plug and on the contact plug, the second portion of the upper separation pattern penetrates through a plurality of upper gate electrodes of the gate electrodes, the vertical memory structure includes an insulating region, a channel layer on a side surface of the insulating region, a first dielectric layer on an external side surface of the channel layer, a data storage layer on an external side surface of the first dielectric layer, a second dielectric layer on an external side surface of the data storage layer, and a pad pattern on the insulating region, the vertical dummy structure includes a dummy insulating region, a dummy channel layer on a side surface of the dummy insulating region, a first dummy dielectric layer on an external side surface of the dummy channel layer, a dummy data storage layer on an external side surface of the first dummy dielectric layer, a second dummy dielectric layer on an external side surface of the dummy data storage layer, and a dummy pad pattern on the dummy insulating region, when viewed on a plane at a first height level that is higher than a height level of a lowermost surface of a lowermost one of the plurality of upper gate electrodes, and lower than a height level of a lowermost surface of the pad pattern, the dummy channel layer of the vertical dummy structure includes a first dummy channel region having a first minimum thickness and a second dummy channel region having a first maximum thickness, when viewed on the plane at the first height level, the channel layer has a substantially uniform thickness, and the first maximum thickness of the second dummy channel region is smaller than the thickness of the channel layer of the vertical memory structure.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) Features will be apparent to those of skill in the art by describing in detail exemplary embodiments with reference to the attached drawings in which:

(2) FIG. **1** is a schematic plan view of a semiconductor device according to example embodiments.

(3) FIGS. **2**, **3**A, **3**B, **4**, **5**, and **6** are schematic diagrams of a semiconductor device according to an example embodiment.

(4) FIGS. **7**A, **7**B, **7**C, and **7**D are schematic plan views of a semiconductor device according to an example embodiment.

(5) FIG. **8** is a schematic plan view of a modified example of a semiconductor device according to an example embodiment.

(6) FIG. **9** is a schematic cross-sectional view of a modified example of a semiconductor device according to an example embodiment.

(7) FIG. **10** is a schematic cross-sectional view of a modified example of a semiconductor device according to an example embodiment.

(8) FIG. **11** is a schematic cross-sectional view of a modified example of a semiconductor device according to an example embodiment.

(9) FIG. **12** is a schematic plan view of a modified example of a semiconductor device according to an example embodiment.

(10) FIG. **13** is a schematic cross-sectional view of a modified example of a semiconductor device according to an example embodiment.

(11) FIGS. **14**A and **14**B are schematic diagrams of a modified example of a semiconductor device according to an example embodiment.

(12) FIGS. **15** and **16** are schematic cross-sectional views of two different modified examples of a semiconductor device according to an example embodiments.

(13) FIG. **17** is a flowchart of a method of fabricating a semiconductor device according to an example embodiment.

(14) FIGS. **18**A, **18**B, **18**C, **19**A, **19**B, **19**C, **20**A, **20**B, **20**C, **21**A, **21**B, **21**C, **22**A, **22**B, and **22**C

are schematic cross-sectional views of stages in a method of fabricating a semiconductor device according to an example embodiment.
(15) FIG. **23** is a schematic diagram of an electronic system including a semiconductor device according to an example embodiment.
(16) FIG. **24** is a schematic perspective view of a data storage system including a semiconductor device according to an example embodiment.
(17) FIG. **25** is a schematic cross-sectional view of a data storage system including a semiconductor device according to an example embodiment.
DETAILED DESCRIPTION
(18) Hereinafter, terms such as "up," "upper portion," "upper surface," "down," "lower portion", "lower surface," "side surface," etc., may be understood as being referred to based on the drawings. Terms such as "upper'", "middle" and "lower" may be replaced with other terms, e.g., "first," "second" and "third," etc. to be used to describe elements of the specification. Terms such as "first" and "second" may be used to describe various elements, but the elements are not limited by the terms, e.g., the terms are merely for differentiation and are not intended to imply or require sequential inclusion, and a "first element" may be referred to as a "second element."
(19) An example of a semiconductor device according to example embodiments will be described with reference to FIGS. **1** to **6**. FIG. **1** is a schematic plan view of a semiconductor device according to example embodiments, and FIG. **2** is a cross-sectional view, taken along line I-I′ of FIG. **1**, of an example of a semiconductor device according to an example embodiment, FIG. **3**A is a partially enlarged view of region "A" of FIG. **2**, FIG. **3**B is a partially enlarged view of region "B" of FIG. **2**, FIG. **4** is a cross-sectional view, taken along line II-II′ of FIG. **1**, of an example of a semiconductor device according to an example embodiment, FIG. **5** is a partially enlarged view of region "C" of FIG. **4**, and FIG. **6** is a cross-sectional view, taken along line III-III′ of FIG. **1**, of an example of a semiconductor device according to an example embodiment.
(20) Referring to FIGS. **1** to **6**, a semiconductor device **1** according to an example embodiment may include a substrate structure **3**, a stack structure ST, vertical memory structures **15***m*, vertical dummy structures **15***d*, and an upper separation pattern **45**.
(21) The semiconductor device **1** may further include separation structures **66**, upper insulating layers **48** and **69**, contact plugs **72**, and bitlines **75**.
(22) The substrate structure **3** may be a semiconductor substrate. The substrate structure **3** may be a silicon substrate. In an implementation, the substrate structure **3** may be a single-crystalline silicon substrate. At least a portion of the substrate structure **3** may be a region doped with an impurity, e.g., a doped region having N-type conductivity type or P-type conductivity type.
(23) The stack structure ST may be on the substrate structure **3**. The stack structure ST may include interlayer insulating layers **8** and gate electrodes **63** alternately and repeatedly stacked in a vertical direction Z, perpendicular to an upper surface of the substrate structure **3**.
(24) The interlayer insulating layers **8**, which may be spaced apart from each other in the vertical direction Z, may include a lowermost interlayer insulating layer **8**L**1**, a next lowermost interlayer insulating layer **8**L**2**, an uppermost interlayer insulating layer **8**U, and middle interlayer insulating layers **8**M between the next lowermost interlayer insulating layer **8**L**2** and the uppermost interlayer insulating layer **8**U. The next lowermost interlayer insulating layer **8**L**2** may have a thickness greater than a thickness of each of the lowermost interlayer insulating layer **8**L**1** and the middle interlayer insulating layers **8**M. The uppermost interlayer insulating layer **8**U may have a thickness greater than a thickness of each of the lowermost interlayer insulating layer **8**L**1** and the middle interlayer insulating layers **8**M.
(25) The gate electrodes **63** may include a lower gate electrode **63**L, upper gate electrodes **63**U**1**, **63**U**2**, and **63**U**3**, and middle gate electrodes **63**M between the lower gate electrode **63**L and the upper gate electrodes **63**U**1**, **63**U**2**, and **63**U**3**.
(26) The lower gate electrode **63**L may be a lower select gate electrode. At least some of the upper

gate electrodes **63**U**1**, **63**U**2**, and **63**U**3** may be upper select gate electrodes. At least a plurality of the middle gate electrodes **63**M may be wordlines.

(27) The upper gate electrodes **63**U**1**, **63**U**2**, and **63**U**3** may include a first upper gate electrode **63**U**1** in or at an uppermost portion, a second upper gate electrode **63**U**2** below the first upper gate electrode **63**U**1**, and a third upper gate electrode **63**U**3** below the second upper gate electrode **63**U**2**.

(28) The interlayer insulating layers **8** may be formed of silicon oxide. The gate electrodes **63** may include, e.g., a doped silicon layer, a metal layer, a metal nitride layer, or a metal-semiconductor compound layer. In an implementation, the gate electrodes **63** may include, e.g., doped silicon, tungsten (W), ruthenium (Ru), molybdenum (Mo), nickel (Ni), nickel silicon (NiSi), cobalt (Co), cobalt silicon (CoSi), titanium (Ti), titanium nitride (TiN), or tungsten nitride (WN). As used herein, the term “or” is not an exclusive term, e.g., “A or B” would include A, B, or A and B.

(29) The vertical memory structure **15***m* may include a plurality of vertical memory structures **15***m*. Hereinafter, one of the plurality of vertical memory structures **15***m* will be mainly described.

(30) The vertical memory structure **15***m* may penetrate through the stack structure ST. The vertical memory structure **15***m* may penetrate through the stack structure ST and may be in contact with the substrate structure **3**.

(31) The vertical memory structure **15***m* may include an insulating region **21***m*, a channel layer **24***m* on a side surface of the insulating region **21***m*, a dielectric structure **27***m* on an external side surface of the channel layer **24***m*, and a pad pattern **39***m* on the insulating region **21***m*. The dielectric structure **27***m* may include a first dielectric layer **30***m* on the external side surface of the channel layer **24***m*, a data storage layer **33***m* on an external side surface of the first dielectric layer **30***m*, and a second dielectric layer **36***m* on an external side surface of the data storage layer **33***m*. The data storage layer **33***m* may be between the first dielectric layer **30***m* and the second dielectric layer **36***m*. The first dielectric layer **30***m* may be in contact with the channel layer **24***m*. The pad pattern **39***m* may be on or at a level higher (e.g., farther from the substrate structure **3** in the Z direction) than a level (e.g., distance from the substrate structure **3** in the Z direction) of an uppermost gate electrode (e.g., **63**U**1**) of the gate electrodes **63**.

(32) The channel layer **24***m* may cover a side surface of the pad pattern **39***m*. The channel layer **24***m* may be in contact with the pad pattern **39***m*. The dielectric structure **27***m* may include a portion on or at the same height level as the pad pattern **39***m*.

(33) The channel layer **24***m* may include a silicon layer. The insulating region **21***m* may include silicon oxide. The first dielectric layer **30***m* may include silicon oxide or silicon oxide doped with impurities. The second dielectric layer **36***m* may include silicon oxide or a high-k dielectric material. The data storage layer **33***m* may include a material trapping charges to store data, e.g., silicon nitride. The data storage layer **33***m* may include regions, storing data, in a semiconductor device such as a flash memory device. The pad pattern **39***m* may include doped polysilicon, a metal nitride (e.g., titanium nitride (TiN), or the like), a metal (e.g., tungsten (W), or the like), or a metal-semiconductor compound (e.g., TiSi (titanium silicon), or the like).

(34) The vertical memory structure **15***m* may further include a channel pattern **18***m*.

(35) The channel pattern **18***m* may be in contact with the substrate structure **3**, may penetrate through the lowermost interlayer insulating layer **8**L**1** and the lowermost gate electrode **63**L, and may extend inwardly of or relative to the lower interlayer insulating layer **8**L**2**. The channel pattern **18***m* may include a portion buried in the substrate structure **3**. The channel pattern **18***m* may be formed of an epitaxial material layer epitaxially grown from the substrate structure **3**, e.g., an epitaxial silicon layer. An upper surface of the channel pattern **18***m* may be on a level higher than a level of the lowermost gate electrode **63**L, and may be on a level lower than a level of an upper surface of the lower interlayer insulating layer **8**L**2**.

(36) The dielectric structure **27***m*, the channel layer **24***m*, and the insulating region **21***m* may be on the channel pattern **18***m*. The channel layer **24***m* may cover an external surface of the insulating

region **21***m* and may cover a bottom surface of the insulating region **21***m*. The channel layer **24***m* may be in contact with the channel pattern **18***m*.
(37) The vertical dummy structure **15***d* may include a plurality of the vertical dummy structures **15***d*. In an implementation, a plurality of vertical dummy structures **15***d* may be spaced apart from each other in a first direction Y, parallel to the upper surface of the substrate structure **3**. Hereinafter, one of the plurality of vertical dummy structure **15***d* will be mainly described.
(38) The vertical dummy structure **15***d* may penetrate through the stack structure ST. The vertical dummy structure **15***d* may penetrate through the stack structure ST and be in contact with the substrate structure **3**.
(39) The vertical dummy structure **15***d* may include a dummy insulating region **21***d*, a dummy channel layer **24***d* on a side surface of the dummy insulating region **21***d*, a dummy dielectric structure **27***d* on an external side surface of the dummy channel layer **24***d*, and a dummy pad pattern **39***d* on the dummy insulating region **21***d*. The dummy dielectric structure **27***d* may include a first dummy dielectric layer **30***d* on an external side surface of the dummy channel layer **24***d*, a dummy data storage layer **33***d* on an external side surface of the first dummy dielectric layer **30***d*, and a second dummy dielectric layer **36***d* on an external side surface of the dummy data storage layer **33***d*. The dummy data storage layer **33***d* may be between the first dummy dielectric layer **30***d* and the second dummy dielectric layer **36***d*. The first dummy dielectric layer **30***d* may be in contact with the dummy channel layer **24***d*. The dummy pad pattern **39***d* may be on a level higher than a level of an uppermost gate electrode (e.g., **63**U**1**), among the gate electrodes **63**.
(40) The dummy channel layer **24***d* may cover a side surface of the dummy pad pattern **39***d*. The dummy channel layer **24***d* may be in contact with the dummy pad pattern **39***d*. The dummy dielectric structure **27***d* may include a portion on the same height level as the dummy pad pattern **39***d*.
(41) The vertical dummy structure **15***d* may further include a dummy channel pattern **18***d*. The dummy channel pattern **18***d* may be in contact with the substrate structure **3**, may penetrate through a lowermost interlayer insulating layer **8**L**1** and a lowermost gate electrode **63**L, and may extend inwardly of or with respect to a lowermost interlayer insulating layer **8**L**2**.
(42) In an implementation, the vertical dummy structure **15***d* may be formed at the same time as the vertical memory structure **15***m*, and may include the same material layers as the vertical memory structure **15***m*. The dummy insulating region **21***d*, the dummy channel layer **24***d*, the dummy dielectric structure **27***d*, the dummy pad pattern **39***d*, and the dummy channel pattern **18***d* of the vertical dummy structure **15***d* may correspond to the insulating region **21***m*, the channel layer **24***m*, the dielectric structure **27***m*, the pad pattern **39***m*, and the channel pattern **18***m* of the vertical memory structure **15***m*, respectively. In an implementation, the dummy insulating region **21***d* may be formed of the same material as the insulating region **21***m*, the dummy channel layer **24***d* may be formed of the same material as the channel layer **24***m*, the dummy dielectric structure **27***d* may be formed of the same material as the dielectric structure **27***m*, the dummy pad pattern **39***d* may be formed of the same material as the pad pattern **39***m*, and the dummy channel pattern **18***d* may be formed of the same material as the channel pattern **18***m*.
(43) The semiconductor device **1** may further include a first oxide layer **57***a* in contact with the channel pattern **18***m* between the lowermost gate electrode **63**L and the channel pattern **18***m*, and a second oxide layer **57***b* in contact with the dummy channel pattern **18***d* between the lowermost gate electrode **63**L and the dummy channel pattern **18***d*.
(44) The upper separation pattern **45** may extend (e.g., lengthwise) in the first direction Y, parallel to the upper surface of the substrate structure **3**, and may include a first portion **45***a* intersecting the vertical dummy structure **15***d* and a second portion **45***b* extending from the first portion **45***a* and penetrating through a portion of the stack structure ST. The upper separation pattern **45** may further include an upper portion **45***c* covering the upper surface of the stack structure ST. In the upper separation pattern **45**, the first portion **45***a*, the second portion **45***b*, and the upper portion **45***c* may

be integrated with each other (e.g., may have a one-piece, monolithic structure).
(45) The second portion **45***b* of the upper separation pattern **45** may penetrate through the upper gate electrodes **63**U**1**, **63**U**2**, and **63**U**3**. In an implementation, as illustrated in FIGS. **4** and **5**, the second portion **45***b* of the upper separation pattern **45** may penetrate through the three upper gate electrodes **63**U**1**, **63**U**2**, and **63**U**3**. In an implementation, the second portion **45***b* of the upper separation pattern **45** may penetrate through more than three upper gate electrodes.
(46) The semiconductor device **1** may further include a gate dielectric layer **60** covering an upper surface and a lower surface of each of the gate electrodes **63** and between the vertical memory structure **15***m* and the gate electrodes **63**, between the vertical dummy structure **15***d* and the gate electrodes **63**, and between the upper gate electrodes **63**U**1**, **63**U**2**, and **63**U**3** and the upper separation pattern **45**. The gate dielectric layer **60** may include, e.g., silicon oxide or a high-k dielectric material.
(47) The upper insulating layers **48** and **69** may include a first upper insulating layer **48** on the upper portion **45***c* of the upper separation pattern **45** and a second upper insulating layer **69** on the first upper insulating layer **48**.
(48) The separation structures **66** may penetrate through the first upper insulating layer **48** and the stack structure ST. When viewed in a plan view, the separation structures **66** may have a linear shape extending in the first direction Y.
(49) Each of the separation structures **66** may include a separation pattern **66***b* and a separation spacers **66***a* on side surfaces of the separation pattern **66***b*. The separation spacer **66***a* may be formed of an insulating material. The separation pattern **66***b* may be formed of a conductive material. In an implementation, the separation pattern **66***b* may be formed of an insulating material.
(50) The contact plug **72** may penetrate through the first and second upper insulating layers **48** and **69** and the upper portion **45***c*, and may be in contact with the pad pattern **39***m* of the vertical memory structure **15***m*.
(51) The bitline **75** may be electrically connected to the contact plug **72** on the second upper insulating layer **69**. The bitline **75** may have a linear shape, parallel to an upper surface of the substrate structure **3**, and may extend in a second direction X, perpendicular to the first direction Y.
(52) The data storage layer **33***m* and the dummy data storage layer **33***d* may have substantially the same thickness.
(53) The second dielectric layer **30***m* and the first dummy dielectric layer **30***d* may have substantially the same thickness.
(54) The first dielectric layer **36***m* and the second dummy dielectric layer **36***d* may have substantially the same thickness, e.g., as measured at or on the same height level as at least one of the middle gate electrodes **63**M.
(55) The channel layer **24***m* and the dummy channel layer **24***d* may have substantially the same thickness, e.g., as measured at or on the same height level as at least one of the middle gate electrodes **63**M.
(56) The dummy data storage layer **33***d* may be divided into dummy data storage portions spaced apart from each other in the second direction X by the upper separation pattern **45**, at any one height level that is higher (e.g., farther from the substrate structure **3** in the Z direction) than that of a lower end of the upper separation pattern **45**.
(57) A portion of the dummy pad pattern **39***d* may be divided into pad portions, spaced apart from each other in the second direction X by the first portion **45***a* of the upper separation pattern **45**, on any one height level, higher than that of a lower end of the upper separation pattern **45**.
(58) The dummy channel layer **24***d* may include a portion having a thickness different from a thickness of the channel layer **24***m*, as measured at or on a height level that is higher than a level of the lower end of the upper separation pattern **45**. In an implementation, the channel layer **24***m* and the dummy channel layer **24***d* may have different thicknesses at the same height level as at least one of the upper gate electrodes **63**U**1**, **63**U**2**, and **63**U**3**. In an implementation, the dummy channel

layer **24***d* may have a thickness smaller than that of the channel layer **24***m* at the same height level as at least one of the upper gate electrodes **63**U**1**, **63**U**2**, and **63**U**3**.
(59) The dummy channel layer **24***d* may include a portion of which thickness is decreased at a point along the vertical direction Z from the lower end of the upper separation pattern **45** toward the upper end of the upper separation pattern **45**. The channel layer **24***m* may have a substantially uniform thickness at a point along the vertical direction Z from the lower end of the upper separation pattern **45** toward the upper end of the upper separation pattern **45**.
(60) The second dummy dielectric layer **36***d* may include a portion having a thickness different from a thickness of the first dielectric layer **36***m* at a height level that is higher than a height level of the lower end of the upper separation pattern **45**. In an implementation, the second dummy dielectric layer **36***d* and the first dielectric layer **36***m* may have different thicknesses at the same height level as at least one of the upper gate electrodes **63**U**1**, **63**U**2**, and **63**U**3**. In an implementation, the thickness of the second dummy dielectric layer **36***d* may be greater than the thickness of the first dielectric layer **36***m*.
(61) The second dummy dielectric layer **36***d* may include a portion of which thickness is increased at different points along the vertical direction Z from the lower end of the upper separation pattern **45** toward the upper end of the upper separation pattern **45**. The first dielectric layer **36***m* may have a substantially uniform thickness at different points along the vertical direction Z from the lower end of the upper separation pattern **45** toward the upper end of the upper separation pattern **45**.
(62) The dummy channel layer **24***d* may include dummy channel regions having different thicknesses and the second dummy dielectric layer **36***d* may include dummy dielectric regions having different thicknesses at a height level, higher than the lower end of the upper separation pattern **45**.
(63) Next, planar shapes at different height levels of the semiconductor device according to an example embodiment will be described with reference to FIGS. **7**A, **7**B, **7**C and **7**D. FIGS. **7**A, **7**B, **7**C, and **7**D are plan views of planar shapes at different height levels in region “D” of FIG. **1**. For example, FIG. **7**A is a schematic plan view of region “D” of FIG. **1** at a height level P of the first portion **45***a* of the upper separation pattern **45**, FIG. **7**B is a schematic plan view of region “D” of FIG. **1** at a first height level L**1** of the first upper gate electrode **63**U**1**, FIG. **7**C is a schematic plan view of region “D” of FIG. **1** at the second height level L**2** of a third upper gate electrode **63**U**3**, and FIG. **7**D is a schematic plan view of region “D” of FIG. **1** at the third height level L**3** of a middle gate electrode **63**M.
(64) Referring to FIG. **7**A, when viewed in plan view, the vertical dummy structure **15***d* may be divided by the first portion **45***a* of the upper separation pattern **45** at a height level of the first portion **45***a* of the upper separation pattern **45**, e.g., at a height level P between the upper end and the lower end of the first portion **45***a*. A maximum width of the first portion **45***a* in the second direction X may be greater than a maximum width of the second portion **45***b* in the second direction X, at the height level P.
(65) When viewed in plan view, the dummy channel layer **24***d* may include dummy channel regions **24***d*_u**1** and **24***d*_u**2** having different thicknesses at the height level P. Among the dummy channel regions **24***d*_u**1** and **24***d*_u**2** having different thicknesses, a dummy channel region **24***d*_u**1** having a relatively lower thickness may be adjacent to the first portion **45***a*.
(66) When viewed in plan view, the dummy channel layer **24***d* may have a thickness smaller than that of the channel layer **24***m* at the height level P.
(67) When viewed in plan view, the second dummy dielectric layer **36***d* may have a thickness greater than that of the first dielectric layer **36***m* at the height level P.
(68) When viewed in plan view, the dummy data storage layer **33***d* may have substantially the same thickness as the data storage layer **33***m* at the height level P.
(69) Referring to FIG. **7**B, when viewed on a plane at a first height level L**1** higher than the height level of the lower end of the upper separation pattern **45**, the upper separation pattern **45** may

divide each of the dummy data storage layer **33***d* and the first dummy dielectric layer **30***d*. In an implementation, the dummy data storage layer **33***d* may be divided into dummy data storage portions spaced apart from each other in the second direction X by the first portion **45***a* of the upper separation pattern **45**.
(70) In an implementation, the first height level L**1** may be the same as the height level of the first upper gate electrode **63**U**1**. In an implementation, the first height level L**1** may be lower than the height level of the first upper gate electrode **63**U**1** or higher than the height level of the first upper gate electrode **63**U**1**.
(71) When viewed on a plane at the first height level L**1**, the gate dielectric layer **60** may be between the vertical memory structure **15***m* and the first upper gate electrode **63**U**1**, between the vertical dummy structure **15***d* and the first upper gate electrode **63**U**1**, and between the upper separation pattern **45** and the first upper gate electrode **63**U**1**.
(72) When viewed on the plane at the first height level L**1**, the first dummy data storage layer **33***d* may have a substantially uniform thickness.
(73) When viewed on the plane on the first height level L**1**, the first dummy data storage layer **33***d* and the data storage layer **33***m* may have substantially the same thickness.
(74) When viewed on the plane at the first height level L**1**, the dummy channel layer **24***d* may include a first dummy channel region **24***d*_L**1***a* and a second dummy channel region **24***d*_L**1***b* having different thicknesses (e.g., in a region facing or radially aligned with the dummy data storage layer **33***d*). A thickness t**2** of the first dummy channel region **24***d*_L**1***a* may be smaller than a thickness t**1** of the second dummy channel region **24***d*_L**1***b*. A distance between the first dummy channel region **24***d*_L**1***a* and the upper separation pattern **45** may be less than a distance between the second dummy channel region **24***d*_L**1***b* and the upper separation pattern **45**.
(75) When viewed on the plane at the first height level L**1**, the first dummy channel region **24***d*_L**1***a* of the dummy channel layer **24***d* may include a portion having a gradually increased or increasing width.
(76) When viewed on the plane at the first height level L**1**, the second dummy dielectric layer **36***d* may include a first dummy dielectric region **36***d*_L**1***a* between the first dummy channel region **24***d*_L**1***a* and the dummy data storage layer **33***d* and a second dummy dielectric region **36***d*_L**1***b* between the second dummy channel region **24***d*_L**1***b* and the dummy data storage layer **33***d*. A thickness of the first dummy dielectric region **36***d*_L**1***a* may be greater than a thickness of the second dummy dielectric region **36***d*_L**1***b*.
(77) When viewed on the plane at the first height level L**1**, the dummy channel layer **24***d* may further include a dummy channel region **24***d*_L**1***c* facing (e.g., radially aligned with) the upper separation pattern **45**. The dummy channel region **24***d*_L**1***c* may have a smallest thickness t**3** in the dummy channel layer **24***d*. In an implementation, a thickness t**3** of the dummy channel region **24***d*_L**1***c* may be smaller than a thickness t**2** of the first dummy channel region **24***d*_L**1***a* and a thickness t**1** of the second dummy channel region **24***d*_L**1***b*.
(78) When viewed on the plane at the first height level L**1**, the dummy channel layer **24***d* may have a ring shape.
(79) When viewed on the plane at the first height level L**1**, a minimum thickness or a maximum thickness of the dummy channel layer **24***d* having a non-uniform thickness may be smaller than a thickness of the channel layer **24***m* having a substantially uniform thickness. In an implementation, when viewed on a plane at the first height level L**1**, the maximum thickness of the dummy channel layer **24***d*, e.g., the maximum thickness of the second dummy channel region **24***d*_L**1***b* may be smaller than the thickness of the channel layer **24***m*.
(80) When viewed on the plane at the first height level L**1**, a maximum thickness of the second dummy dielectric layer **36***d* may be greater than a thickness of the first dielectric layer **36***m* having a substantially uniform thickness.
(81) When viewed on the plane at the first height level L**1**, the dummy channel layer **24***d* may have

a ring shape and, in the dummy channel layer **24***d*, a thickness t**3** of the dummy channel region **24***d*_L**1***c* at a center of the vertical dummy structure **15***d* in the first direction Y may be smaller than a thickness t**1** of the second dummy channel region **24***d*_L**1***b* at the center of the vertical dummy structure **15***d* in the second direction X.
(82) Referring to FIG. **7**C, when viewed on a plane at a second height level L**2** higher than the lower end of the upper separation pattern **45** and lower than the first height level L**1**, the upper separation pattern **45** may divide each of the dummy data storage layer **33***d* and the first dummy dielectric layer **30***d*. In an implementation, the dummy data storage layer **33***d* may be divided into first and second dummy data storage portions spaced apart from each other in the second direction X by the first portion **45***a* of the upper separation pattern **45**.
(83) In an implementation, the second height level L**2** may be the same as the height level of the third upper gate electrode **63**U**3**. In an implementation, the second height level L**2** may be a height level at a position higher than the height level of the third upper gate electrode **63**U**3** and lower than the height level of the first upper gate electrode **63**U**1**.
(84) When viewed on a plane at the second height level L**2**, the gate dielectric layer **60** may be between the vertical memory structure **15***m* and the third upper gate electrode **63**U**3**, between the vertical dummy structure **15***d* and the third upper gate electrode **63**U**3**, and between the upper separation pattern **45** and the third upper gate electrode **63**U**3**.
(85) When viewed on the plane at the second height level L**2**, the first dummy data storage layer **33***d* may have a substantially uniform thickness.
(86) When viewed on the plane at the second height level L**2**, the first dummy data storage layer **33***d* and the data storage layer **33***m* may have substantially the same thickness.
(87) When viewed on the plane at the second height level L**2**, the dummy channel layer **24***d* may include a third dummy channel region **24***d*_L**2***a* and a fourth dummy channel region **24***d*_L**2***b* having different thicknesses (e.g., in regions facing or radially aligned with the dummy data storage layer **33***d*). A thickness t**5** of the third dummy channel region **24***d*_L**2***a* may be smaller than a thickness t**4** of the fourth dummy channel region **24***d* L**2***b*. A distance between the third dummy channel region **24***d* L**2***a* and the upper separation pattern **45** may be smaller than a distance between the fourth dummy channel region **24***d*_L**2***b* and the upper separation pattern **45**.
(88) When viewed on the plane at the second height level L**2**, in the second dummy dielectric layer **36***d*, a thickness between the third dummy channel region **24***d* L**2***a* and the dummy data storage layer **33***d* may be greater than a thickness between the fourth dummy channel region **24***d*_L**2***b* and the dummy data storage layer **33***d*.
(89) When viewed on the plane at the second height level L**2**, the dummy channel layer **24***d* may further include a dummy channel region **24***d*_L**2***c* facing the upper separation pattern **45**. The dummy channel region **24***d*_L**2***c* may have a smallest thickness in the dummy channel layer **24***d*, e.g., a minimum thickness t**6**. In an implementation, the thickness t**6** of the dummy channel region **24***d*_L**2***c* may be smaller than the thickness t**5** of the third dummy channel region **24***d*L**2***a* and the thickness t**4** of the fourth dummy channel region **24***d*_L**2***b*.
(90) When viewed on the plane at the second height level L**2**, the dummy channel layer **24***d* may have a ring shape.
(91) When viewed on the plane at the second height level L**2**, a minimum thickness or a maximum thickness of the dummy channel layer **24***d* may be smaller than a thickness of the channel layer **24***m* (e.g., which may have a substantially uniform thickness).
(92) When viewed on the plane at the second height level L**2**, a maximum thickness of the second dummy dielectric layer **36***d* may be greater than a thickness of the first dielectric layer **36***m* (which may have a substantially uniform thickness).
(93) The minimum thickness t**6** of the dummy channel layer (**24***d* of FIG. **7**C) at the second height level L**2** may be greater than the minimum thickness t**3** of the dummy channel layer (**24***d* of FIG. **7**B) at the first height level L**1**.

(94) The maximum thickness of the second dummy dielectric layer **36***d* at the second height level L**2** may be smaller than the maximum thickness of the second dummy dielectric layer **36***d* at the first height level L**1**.
(95) When viewed on the plane at the second height level L**2**, the dummy channel layer **24***d* may have a ring shape and, in the dummy channel layer **24***d*, the thickness t**6** of the dummy channel region **24***d*_L**2***c* at the center of the vertical dummy structure **15***d* in the first direction Y may be smaller than the thickness t**4** of the fourth dummy channel region **24***d*_L**2***b* at the center of the vertical dummy structure **15***d* in the second direction X.
(96) Referring to FIG. **7**D, at a height level lower than the height level of the lower end of the upper separation pattern **45**, e.g., at a same height level L**3** as one of the middle gate electrodes **63**M, the dummy channel layer **24***d* and the channel layer **24***m* may have substantially the same thickness, the second dummy dielectric layer **36***d* and the first dielectric layer **36***m* may have substantially the same thickness, and the dummy data storage layer **33***d* and the data storage layer **33***m* may have substantially the same thickness.
(97) Next, a modified example of the semiconductor device according to an example embodiment will be described with reference to FIG. **8**. FIG. **8** is a schematical plan view of region "D" of FIG. **1** at a first height level L**1** of a first upper gate electrode **63**U**1**.
(98) Referring to FIG. **8**, when viewed on the plane at the first height level L**1** as described with reference to FIG. **7**B, the dummy data storage layer **33***d* may include dummy data storage portions spaced apart from each other in the second direction X.
(99) When viewed on the plane at the first height level L**1**, the dummy channel layer **24***d* may include dummy channel portions spaced apart from each other in the second direction X.
(100) When viewed on the plane at the first height level L**1**, each of the dummy channel portions of the dummy channel layer **24***d* may include a first dummy channel region **24***d*_L**1***a*′ having a first thickness t**2**′ and a second dummy channel region **24***d*_L**1***a*′ having a second thickness t**1**′. The second thickness t**1**′ may be greater than the first thickness t**2**′.
(101) When viewed on the plane at the first height level L**1**, the first thickness t**2**′ may be a minimum thickness of the dummy channel layer **24***d*, and the second thickness t**1**′ may be a maximum thickness of the dummy channel layer **24***d.*
(102) When viewed on the plane at the first height level L**1**, at least one of the dummy channel portions of the dummy channel layer **24***d* may include a dummy channel region having a gradually increased width. In an implementation, the first dummy channel region **24***d*_L**1***a*′ may include a portion having a gradually increased or increasing width.
(103) In an implementation, when viewed on the plane at the first height level L**1**, the ring-shaped dummy channel layer **24***d* as illustrated in FIG. **7**B may not be present, but rather may include the dummy channel layer **24***d* including portions spaced apart from each other in the second direction X as illustrated in FIG. **8**. In an implementation, when viewed in a partially enlarged cross-sectional view of FIG. **3**B, in the dummy channel layer **24***d* as illustrated in FIG. **3**B at a level higher than a level of the first height level L**1** and lower than a level of a lower end of the dummy pad pattern **39***d*, a planar shape of the dummy channel layer **24***d* including portions spaced apart from each other in the second direction X may appear as illustrated in FIG. **8**. In an implementation, the dummy channel layer **24***d* described with reference to FIGS. **1** to **6** may have the planar shape described with reference to FIG. **8**, or the planar shape described with reference to FIGS. **7**A to **7**D.
(104) Next, various modified example of the above-described semiconductor device **1** according to an example embodiment will be described. Hereinafter, in description of various modified examples of the above-described semiconductor device **1**, modified or replaced components, among the components of the above-described semiconductor device **1**, will be mainly described and descriptions of components substantially the same as the above-described components or components which can be easily understood from the contents or drawings may be omitted.

(105) FIG. **9** is a cross-sectional view, taken along line I-I′ of FIG. **1**, of a modified example of the semiconductor device **1** according to an example embodiment.
(106) In an implementation, referring to FIG. **9**, the first portion **45***a* of the upper separation pattern **45** as described with reference to FIGS. **2** and **3**B may divide the dummy pad pattern **39***d* into pad portions, spaced apart from each other in the second direction X, while penetrating through the dummy pad pattern **39***d* in the vertical direction Z, and may be replaced with a first portion **45***a′* having a lower surface at a level higher than that of the first upper gate electrode **63**U**1**.
(107) FIG. **10** is a cross-sectional view, taken along line I-I′ of FIG. **1**, of a modified example of the semiconductor device **1** according to an example embodiment.
(108) In an implementation, referring to FIG. **10**, the first portion **45***a* of the upper separation pattern **45** as described with reference to FIGS. **2** and **3**B may divide the dummy pad pattern **39***d* into pad portions, spaced apart from each other in the second direction X, while penetrating through the dummy pad pattern **39***d* in the vertical direction, and may be replaced with a first portion **45***a″* having a lower surface at a level the same or lower than a level of the first upper gate electrode **63**U**1** or the second upper gate electrode **63**U**2**.
(109) FIG. **11** is a cross-sectional view, taken along line II-II′ of FIG. **1**, of a modified example of the semiconductor device **1** according to an example embodiment.
(110) In an implementation, referring to FIG. **11**, the second portion **45***b* of the upper separation pattern **45** as described with reference to FIGS. **4** and **5** may be replaced with a second portion **45***b′* penetrating through a greater number of upper gate electrodes (e.g., **63**U**1**, **63**U**2**, **63**U**3**, **63**U**4**, and **63**U**5**) than three gate electrodes, among the gate electrodes **63**, in the vertical direction Z. Among the upper gate electrodes **63**U**1**, **63**U**2**, **63**U**3**, **63**U**4**, and **63**U**5** spaced apart from each other while being separated from each other in the second direction X by the second portion **45***b* of the upper separation pattern **45**, a single or a plurality of upper gate electrodes **63**U**1** and **63**U**2** in a relatively upper portion may be erase control gate electrodes, and a single or a plurality of upper gate electrodes **63**U**3** and **63**U**4** at a relatively lower portion may be string select gate electrodes.
(111) FIG. **12** is a schematic plan view of region “D” of FIG. **1** at a fourth height level having the same height as a fifth upper gate electrode **63**U**5** at a lowermost level, among the upper gate electrodes **63**U**1**, **63**U**2**, **63**U**3**, **63**U**4**, and **63**U**5** of FIG. **11**.
(112) Referring to FIGS. **11** and **12**, when viewed on a plane at the fourth height level L**4**, the upper separation pattern **45** may be in contact with the first dummy dielectric layer **30***d*, and the dummy data storage layer **33***d* may have a ring shape.
(113) When viewed on the plane at the fourth height level L**4**, the dummy channel layer **24***d* may have a substantially uniform thickness and the second dummy dielectric layer **36***d* may have a substantially uniform thickness.
(114) When viewed on the plane at the fourth height level L**4**, the dummy channel layer **24***d* may have substantially the same thickness as the channel layer **24***m*.
(115) In the above-described embodiments, the semiconductor device **1** may have the planar shape of the vertical dummy structure **15***d* of FIGS. **7**A to **7**D, the planar shape of the vertical dummy structure **15***d* of FIG. **8**, or the planar shape of the vertical dummy structure **15***d* of FIG. **12**.
(116) FIG. **13** is a cross-sectional view, taken along line II-II′ of FIG. **1**, of a semiconductor device **1***a* according to a modified example.
(117) In an implementation, referring to FIG. **13**, the above-described substrate structure (**3** in FIGS. **2** to **6**) may be replaced with a substrate structure **3***a* including a semiconductor substrate **105**, peripheral circuit regions **110**, **115**, and **120** on the semiconductor substrate **105**, and an upper substrate **125** on the peripheral circuit regions **110**, **115**, and **120**.
(118) The semiconductor substrate **105** may be a single-crystalline semiconductor substrate, e.g., a silicon substrate.
(119) The peripheral circuit regions **110**, **115**, and **120** may include a peripheral circuit element **110** on the semiconductor substrate **105**, peripheral interconnections **115** electrically connected to the

peripheral circuit element **110**, and a lower insulating layer **120** covering the peripheral circuit element **110** and the peripheral interconnections **115** on the semiconductor substrate **105**. The peripheral circuit element **110** may include a peripheral transistor including a peripheral gate **110***a*, formed on an active region defined by a device isolation layer **112** on the semiconductor substrate **105**, and a peripheral source/drain **110***b* in the active region.

(120) The upper substrate **125** may include at least one silicon layer. The upper substrate **125** may include a polysilicon layer doped with impurities.

(121) The stack structure ST, the vertical memory structures **15***m*, the vertical dummy structures **15***d*, and the upper separation pattern **45** described above may be on the upper substrate **125**. The vertical memory structures **15***m* and the vertical dummy structures **15***d* may be in contact with the upper substrate **125**.

(122) FIG. **14**A is a cross-sectional view, taken along line I-I′ of FIG. **1**, illustrating a semiconductor device **1***b* according to a modified example, and FIG. **14**B is a partially enlarged view of region “Aa” of FIG. **14**A.

(123) In an implementation, referring to FIGS. **14**A and **14**B, the above-described substrate structure (**3** in FIGS. **2** to **6**) may be replaced with a substrate structure **3***a*′ including a semiconductor substrate **105**, and peripheral circuit regions **110**, **115**, and **120** on the semiconductor substrate **105**, an upper substrate **125** on the peripheral circuit regions **110**, **115**, and **120**, and horizontal layers **130** and **135** on the upper substrate **125**. The semiconductor substrate **105** and the peripheral circuit regions **110**, **115**, and **120** may be substantially the same as the semiconductor substrate **105** and the peripheral circuit regions **110**, **115**, and **120** described with reference to FIG. **13**.

(124) The horizontal layers **130** and **135** may include a first horizontal layer **130** and a second horizontal layer **135** on the first horizontal layer **135**. The first and second horizontal layers **130** and **135** may be formed of a polysilicon layer having N-type conductivity. At least a portion of the upper substrate **125** may be formed of a polysilicon layer having N-type conductivity.

(125) In the above-described stack structure ST, the gate electrodes **63** may include a plurality of lower gate electrodes **63**L**1** and **63**L**2** spaced apart from each other in the vertical direction Z, and the interlayer insulating layers **8** may include a middle interlayer insulating layers **8**M on a lowermost interlayer insulating layer **8**L.

(126) Among the plurality of lower gate electrodes **63**L**1** and **63**L**2**, a lower gate electrode **63**L**1** at a relatively lower portion may be a lower erase control gate electrode, and a lower gate electrode **63**L**2** at a relatively upper portion may be a lower select gate electrode.

(127) In the lower region as illustrated in FIG. **14**B, the above-described vertical memory structure **15***m* may be modified into a vertical memory structure **15***m*′ including an insulating region **21***m* extending inwardly of the upper substrate **125**, a channel layer **24***m* covering a side surface and a bottom surface of the insulating region **21***m*, and a dielectric structure **27***m* covering an external side surface and a bottom surface of the channel layer **24***m*.

(128) In the lower region as illustrated in FIG. **14**B, a vertical dummy structure **15***d*′ may include a dummy insulating region **21***d* extending inwardly of the upper substrate **125**, a dummy channel layer **24***d* covering an external side surface and a bottom surface of the dummy insulating region **21***d*, and a dummy dielectric structure **27***d* covering an external side surface and a bottom surface of the dummy channel layer **24***d*.

(129) The first horizontal layer **130** may penetrate through the dielectric structure **27***m* and be in contact with the channel layer **24***m*, and may penetrate through the dummy dielectric structure **27***d* and be in contact with the dummy channel layer **24***d*.

(130) FIG. **15** is a cross-sectional view, taken along line I-I′ of FIG. **1**, illustrating a semiconductor device **1***c* according to a modified example.

(131) In an implementation, referring to FIG. **15**, the semiconductor device **1***c* according to a modified example may include a lower semiconductor chip LC and an upper semiconductor chip

UC bonded to the lower semiconductor chip LC.
(132) The lower semiconductor chip LC may include the semiconductor device **1** described with reference to FIGS. **1** to **11**. In an implementation, the lower semiconductor chip LC may include the substrate structure **3**, the stack structure ST, the vertical memory structures **15***m*, the vertical dummy structures **15***d*, and the upper separation pattern **45**, the separation structures **66**, the upper insulating layers **48** and **69**, the contact plugs **72**, and the bitlines **75**.
(133) The lower semiconductor chip LC may include a lower insulating layer **80**, covering the bitlines **75** on the upper insulating layers **48** and **69**, and a lower bonding pad **85** having an upper surface coplanar with an upper surface of the lower insulating layer **80** and buried in the lower insulating layer **80**.
(134) The upper semiconductor chip UC may include a semiconductor substrate **205**, peripheral circuit regions **210**, **215**, and **220** below the semiconductor substrate **205**, and an upper bonding pad **225** below the peripheral circuit region **210**, **215**, and **220**.
(135) The peripheral circuit regions **210**, **215**, and **220** may include a peripheral circuit element **210** below the semiconductor substrate **205**, peripheral interconnections **215** electrically connected to the peripheral circuit element **210**, and an upper insulating layer **220** covering the peripheral circuit element **210** and the peripheral interconnections **215** below the semiconductor substrate **205**. The peripheral circuit device **210** may include a peripheral transistor including a peripheral gate **210***a*, below an active region defined by a device isolation layer **212** in the semiconductor substrate **205**, and a peripheral source/drain **210***b* in the active region. The peripheral circuit regions **210**, **215**, and **220** may be between the semiconductor substrate **205** and the stack structure ST.
(136) The upper bonding pad **225** may be buried in the upper insulating layer **220** and may be bonded to the lower bonding pad **85**.
(137) The lower and upper bonding pads **85** and **225** may be formed of conductive materials, which may be bonded to each other, e.g., copper (Cu).
(138) FIG. **16** is a cross-sectional view, taken along line I-I′ of FIG. **1**, of a semiconductor device **1***d* according to a modified example.
(139) In an implementation, referring to FIG. **16**, the semiconductor device **1***d* according to the modified example may include a lower semiconductor chip LC′ and an upper semiconductor chip UC′ bonded to the lower semiconductor chip LC′.
(140) The upper semiconductor chip UC′ may be substantially the same as the upper semiconductor chip UC described with reference to FIG. **15**.
(141) The lower semiconductor chip LC′ may include the upper substrate **125**, the horizontal layers **130** and **135**, the stack structure ST, the vertical memory structure **15***m*′, and the vertical dummy structure **15***d*′, which are the same as described with reference to FIGS. **14**A and **14**B, and may include the upper separation pattern **45**, the separation structures **66**, the upper insulating layers **48** and **69**, the contact plugs **72**, and the bitlines **75**, which are the same as described with reference to FIGS. **1** to **11**.
(142) The lower semiconductor chip LC′ may further include the lower insulating layer **80** and the lower bonding pad **85** described with reference to FIG. **15**. The lower and upper bonding pads **85** and **225** may be bonded to each other as described with reference to FIG. **15**.
(143) Next, an example of a method of fabricating a semiconductor device according to an example embodiment will be described with reference to FIGS. **17** to **22**C. FIG. **17** is a flowchart of a method of fabricating a semiconductor device according to an example embodiment, FIGS. **18**A, **19**A, **20**A, **21**A and **22**A are cross-sectional views taken along line I-I′ of FIG. **1**, FIG. **18**B, FIG. **19**B, FIG. **20**B, FIG. **21**B, and FIG. **22**B are cross-sectional views taken along line II-II′ of FIG. **1**, and FIGS. **18**C, FIG. **19**C, FIG. **20**C, FIG. **21**C, and FIG. **22**C are cross-sectional views taken along lines III-III′ of FIG. **1**.
(144) Referring to FIGS. **17**, **18**A, **18**B, and **18**C, a substrate structure **3** may be formed. The substrate structure **3** may be a semiconductor substrate. In an implementation, the substrate

structure **3** may be formed by being replaced with the substrate structure (**3***a* of FIG. **13**) as described with reference to FIG. **13**. The substrate structure **3** may be a silicon substrate. At least a portion of the substrate structure **3** may be a region doped with impurities, e.g., a doped region having N-type conductivity type and/or P-type conductivity type.

(145) In operation S**10**, preliminary stack structures **8** and **10** including interlayer insulating layers **8** and mold layers **10** may be formed. The preliminary stack structures **8** and **10** may be formed on the substrate structure **3**.

(146) The interlayer insulating layers **8** may be formed of silicon oxide. The mold layers **10** may be formed of a material different from that of the interlayer insulating layers **8**. In an implementation, the mold layers **10** may be formed of silicon nitride or polysilicon.

(147) The interlayer insulating layers **8** may include a lowermost interlayer insulating layer **8**L**1**, a next lowermost interlayer insulating layer **8**L**2**, an uppermost interlayer insulating layer **8**U, and middle interlayer insulating layer **8**M between the next lowermost interlayer insulating layer **8**L**2** and the uppermost insulating layer **8**U. The next lowermost interlayer insulating layer **8**L**2** may be formed to have a thickness greater than a thickness of the lowermost interlayer insulating layer **8**L**1** and a thickness of the middle interlayer insulating layers **8**M. The uppermost interlayer insulating layer **8**U may be formed to have a thickness greater than the thickness of the lowermost interlayer insulating layer **8**L**1** and the thickness of the middle interlayer insulating layers **8**M.

(148) The mold layers **10** may include a lower mold layer **10**L, a plurality of upper mold layers **10**U**1**, **10**U**2**, and **10**U**3**, and a middle mold layer **10**M between the lower mold layer **10**L and the plurality of upper mold layers **10**U**1**, **10**U**2**, and **10**U**3**.

(149) In operation S**15**, vertical memory structures **15***m* and vertical dummy structures **15***d* may be formed to penetrate through the preliminary stack structures **8** and **10**.

(150) Forming the vertical memory structures **15***m* and the vertical dummy structures **15***d* may include forming holes **12** to penetrate through the preliminary stack structures **8** and **10** and to expose the substrate structure **3** and simultaneously forming the vertical memory structures **15***m* and the vertical dummy structures **15***d* in the holes. In an implementation, forming each of the vertical memory structures **15***m* may include forming a channel pattern **18***m* epitaxially grown from the substrate structure **3** exposed by the hole **12**, forming a dielectric structure **27***m* on a sidewall of the hole **12** on the channel pattern **18***m*, forming a channel layer **24***m* to cover the dielectric structure **27***m* and to be in contact with the channel pattern **18***m*, forming an insulating region **21***m* to fill a portion of the hole **12** on the channel layer **24***m*, and forming a pad pattern **39***m* to fill a remaining portion of the hole **12** on the insulating region **21***m*. Forming each of the vertical dummy structure **15***d* may include forming a dummy channel pattern **18***d* epitaxially grown from the substrate structure **3** exposed by the hole **12**, forming a dummy dielectric structure **27***d* on a sidewall of the hole **12** on the dummy channel pattern **18***d*, forming a dummy channel layer **24***d* to cover the dummy dielectric structure **27***d* and to be in contact with the dummy channel pattern **18***d*, forming a dummy insulating region **21***d* to fill a portion of the hole **12** on the dummy channel layer **24***d*, and forming a dummy pad pattern **39***d* to fill a remaining portion of the hole **12** on the dummy insulating region **21***d*.

(151) Forming the dielectric structure **27***m* may include sequentially forming a second dielectric layer **36***m*, the data storage layer **33***m*, and the first dielectric layer **30***m*. Forming the dummy dielectric structure **27***d* may include sequentially forming a second dummy dielectric layer **36***d*, the dummy data storage layer **33***d*, and the first dummy dielectric layer **30***d*.

(152) Referring to FIGS. **17**, **19**A, **19**B, and **19**C, in operation S**20**, a groove **42** may be formed to penetrate through a portion of the preliminary structures **8** and **10** and to intersect the vertical dummy structures **15***d*. The groove **42** may have a linear shape extending in the Y direction. The grooves **42** may intersect an upper portion of each of the vertical dummy structures **15***d* while penetrating through a portion of each of the vertical dummy structures **15***d*. A width of the groove **42** in an X direction may be smaller than a width of each of the vertical dummy structure **15***d* in the

X direction.
(153) The groove **42** may penetrate through the upper mold layers **10***u***1**, **10***u***2**, and **10***u***3**, among the mold layers **10**.
(154) In an implementation, the grooves **42** may penetrate through a portion of the dummy pad pattern **39***d* of each of the vertical dummy structures **15***d*.
(155) In an implementation, the groove **42** may penetrate through the entire dummy pad pattern **39***d* of each of the vertical dummy structures **15***d*.
(156) Referring to FIGS. **17**, **20**A, **20**B, and **20**C, in operation S**25**, an upper separation pattern **45** may be formed to fill at least the groove **42**. The upper separation pattern **45** may fill the groove **42** and may cover upper surfaces of the preliminary laminated structures **8** and **10**. The upper separation pattern **45** may be formed of silicon oxide.
(157) In the upper separation pattern **45**, the portions intersecting the vertical dummy structures **15***d* may be referred to as a first portion **45***a*, a portion penetrating through a portion of the preliminary lamination structures **8** and **10** may be referred to as a second portion **45***b*, and a portion covering the upper surfaces of the preliminary laminated structures **8** and **10** may be referred to as an upper portion **45***c*.
(158) In the upper separation pattern **45**, the first portion **45***a* intersecting the vertical dummy structures **15***d* may penetrate through a portion of the dummy data storage layer **33***d* of each of the vertical dummy structures **15***d* to divide a portion of the dummy data storage layer **33***d*.
(159) A first upper insulating layer **48** may be formed on the upper separation pattern **45**. The first upper insulating layer **48** may be formed of silicon oxide harder than the silicon oxide of the upper separation pattern **45**. In an implementation, the first upper insulating layer **48** may be formed of silicon oxide formed at higher temperatures than the silicon oxide of the upper separation pattern **45**.
(160) In operation S**30**, separation trenches **51** may be formed. The separation trenches **51** may penetrate through the first upper insulating layer **48**, the upper portion **45***c* of the upper separation pattern **45**, and the preliminary laminated structures **8** and **10**. The separation trenches **51** may expose the mold layers **10** of the preliminary laminated structures **8** and **10**.
(161) Referring to FIGS. **17**, **21**A, **21**B and FIG. **21**C, in operation S**35**, the mold layers (**10** of FIGS. **19**A to **19**C) exposed by the separation trenches **51** may be removed.
(162) As the mold layers (**10** of FIGS. **19**A to **19**C) are removed, empty space **54** may be formed to expose the vertical memory structures **15***m*, the vertical dummy structures **15***d*, and the upper separation pattern **45**.
(163) In an implementation, the vertical memory structures **15***m* and the vertical dummy structures **15***d* may serve as a support to help prevent the interlayer insulating layers from being bent or deformed by the empty spaces **54** formed while removing the mold layers (**10** of FIG. **19**A to **19**C). Accordingly, the vertical dummy structures **15***d* may also be referred to as vertical support structures.
(164) In operation S**40**, an oxidation process may be performed. The oxidation process may be a thermal oxidation process.
(165) An exposed surface of the channel pattern **18***m* may be oxidized by the oxidation process to form a first oxide layer **57***a*, and a surface of the dummy channel pattern **18***d* may be oxidized by the oxidation process to form a second oxide layer **57***b*.
(166) Oxygen used in the oxidation process may permeate through the upper separation pattern **45** exposed by the empty spaces **54**, and a portion of the dummy channel layer **24***d* of the vertical dummy structures **15***d* may be oxidized by the oxidation process. Accordingly, the dummy channel layer **24***d* may be oxidized to have various thicknesses, as described in FIGS. **7**A to **7**D. The dummy channel layer **24***d* may be oxidized to have a thickness as in FIG. **8** by the oxidation process.
(167) The oxidation process may cure a surface of the second dielectric layer **36***m*. In an

implementation, as the mold layers (**10** of FIGS. **19**A to **19**C) are removed, the surface of the second dielectric layer **36***m* could be damaged or the mold layers (**10** of FIGS. **19**A to **19**C) may finely remain. The oxidation process may cure the damaged surface of the second dielectric layer **36***m*, or may oxidize the finely remaining mold layers (**10** of FIGS. **19**A to **19**C) to completely remove the mold layers (**10** of FIGS. **19**A to **19**C).
(168) The oxidation process may help improve interfacial characteristics between the second dielectric layer **36***m* and the data storage layer **33***m*. Accordingly, data storage characteristics of the semiconductor device may be improved.
(169) Referring to FIGS. **17**, **22**A, **22**B, and **22**C, in operation S**45**, gate layers may be formed. The forming of the gate layers may include forming a gate dielectric layer **60** to conformally cover internal walls of the empty spaces **54** and sequentially forming gate electrodes **63** to fill the empty spaces **54**.
(170) In operation S**50**, separation structures **63** may be formed. The forming of the separation structures **63** may include forming separation spacers **66***a* on side surfaces of the separation trenches **51** and forming separation patterns **66***b* to fill the separation trenches **51**.
(171) Returning to FIGS. **1** to **6** together with FIG. **17**, a second upper insulating layer **69** may be formed on the separation structures **63** and the first upper insulating layer **48**. In operation S**55**, contact plugs **72** may be formed. The contact plugs **72** may penetrate through the first and second upper insulating layers **48** and **69** and the upper portion **45***c* of the upper separation pattern **45**, and may be in contact with the pad patterns **39***m* of the vertical memory structures **15***m*. In operation S**60**, an interconnection may be formed. The interconnection may include bitlines **75**. The bitlines **75** may be electrically connected to the contact plugs **72**.
(172) In the above-described embodiments, to help improve characteristics of the vertical memory structures **15***m*, the oxidation process (**40** in FIG. **17**) may be performed immediately after operation S**20**, in which grooves (**42** of FIGS. **19**A to **19**C) are formed to penetrate a portion of the preliminary stack structure and to intersect the vertical dummy structures **15***d*.
(173) In the above-described embodiments, to help improve characteristics of the vertical memory structures **15***m*, the oxidation process (**40** in FIG. **17**) may be performed immediately after operation S**25**, in which the upper separation patterns (**45** of FIGS. **20**A to **20**C) are formed to fill the groove (**42** of FIGS. **19**A to **19**C).
(174) Next, data storage systems including a semiconductor device according to an example embodiment will be described with reference to FIGS. **23**, **24** and **25**, respectively.
(175) FIG. **23** is a schematic diagram of an electronic system including a semiconductor device according to an example embodiment.
(176) Referring to FIG. **23**, a data storage system **1000** according to an example embodiment may include a semiconductor device **1100** and a controller **1200** electrically connected to the semiconductor device **1100**. The data storage system **1000** may be implemented by a storage device including the semiconductor device **1100** or an electronic device including a storage device. In an implementation, the data storage system **1000** may be implemented by a solid state drive device (SSD), a universal serial bus (USB), a computing system, a medical device, or a communications device, including the semiconductor device **1100**.
(177) In an implementation, the data storage system **1000** may be implemented by an electronic system storing data.
(178) The semiconductor device **1100** may be implemented by a semiconductor device described in the above-described example embodiments with reference to FIGS. **1** to **22**C. The semiconductor device **1100** may include a first structure **1100**F and a second structure **1100**S on the first structure **1100**F.
(179) The first structure **1100**F may be a peripheral circuit structure including a decoder circuit **1110**, a page buffer **1120**, and a logic circuit **1130**.
(180) The second structure **1100**S may be a memory cell structure including a bitline BL, a

common source line CSL, wordlines WL, first and second upper gate lines UL**1** and UL**2**, first and second gate lower lines LL**1** and LL**2**, and memory cell strings CSTR disposed between the bitline BL and the common source line CSL.
(181) In the second structure **1100**S, each of the memory cell strings CSTR may include lower transistors LT**1** and LT**2** adjacent to the common source line CSL, upper transistors UT**1** and UT**2** adjacent to the bitline BL, and a plurality of memory cell transistors MCT between the lower transistors LT**1** and LT**2** and the upper transistors UT**1** and UT**2**. The number of the lower transistors LT**1** and LT**2** and the number of the upper transistors UT**1** and UT**2** may vary according to example embodiments.
(182) In an implementation, the upper transistors UT**1** and UT**2** may include a string select transistor, and the lower transistors LT**1** and LT**2** may include a ground select transistor. The lower gate lines LL**1** and LL**2** may be gate electrodes of lower transistors LT**1** and LT**2**, respectively. The wordlines WL may be gate electrodes of memory cell transistors MCT, and the gate upper lines UL**1** and UL**2** may be gate electrodes of the upper transistors UT**1** and UT**2**, respectively.
(183) The above-described gate electrodes **63** may constitute the gate lower lines LL**1** and LL**2**, the wordlines WL, and the gate upper lines UL**1** and UL**2**.
(184) In an implementation, the lower transistors LT**1** and LT**2** may include a lower erase control transistor LT**1** and a ground select transistor LT**2** connected to each other in series. The upper transistors UT**1** and UT**2** may include a string select transistor UT**1** and an upper erase control transistor UT**2** connected to each other in series. At least one of the lower erase control transistor LT**1** and the upper erase control transistor UT**1** may be used for an erase operation to erase data stored in the memory cell transistors MCT based on gate induced drain leakage (GIDL).
(185) The common source line CSL, the first and second gate lower lines LL**1** and LL**2**, word lines WL, and the first and second gate upper lines UL**1** and UL**2** may be electrically connected to the decoder circuit **1110** through first connection interconnections **1115** extending from the first structure **1100**F to the second structure **1100**S.
(186) The bitlines BL may be electrically connected to the page buffer **1120** through second connection interconnections **1125** extending from the first structure **1100**F to the second structure **1100**S. The bitlines BL may be the above-described bitlines **75**.
(187) In the first structure **1100**F, the decoder circuit **1110** and the page buffer **1120** may perform a control operation on at least one select memory cell transistor of the plurality of memory cell transistors MCT. The decoder circuit **1110** and the page buffer **1120** may be controlled by a logic circuit **1130**.
(188) The semiconductor device **1100** may further include an input/output pad **1101**. The semiconductor device **1100** may communicate with the controller **1200** through the input/output pad **1101** electrically connected to the logic circuit **1130**. The input/output pad **1101** may be electrically connected to the logic circuit **1130** through an input/output connection line **1135** extending from the first structure **1100**F to the second structure **1100**S. Accordingly, the controller **1200** may be electrically connected to the semiconductor device **1100** through the input/output pad **1101** and may control the semiconductor device **1100**.
(189) The controller **1200** may include a processor **1210**, a NAND controller **1220**, and a host interface **1230**. In an implementation, the data storage system **1000** may include a plurality of semiconductor devices **1100**. In this case, the controller **1200** may control the plurality of semiconductor devices **1100**.
(190) The processor **1210** may control overall operation of the data storage system **1000** including the controller **1200**. The processor **1210** may operate according to predetermined firmware and may control the NAND controller **1220** to access the semiconductor device **1100**. The NAND controller **1220** may include a NAND interface **1221** for processing communication with the semiconductor device **1100**. A control command for controlling the semiconductor device **1100**, data to be written in the memory cell transistors MCT of the semiconductor device **1100**, and data

to be read from the memory cell transistors MCT of the semiconductor device **1100** may be transmitted through the NAND interface **1221**. The host interface **1230** may provide a communications function between the data storage system **1000** and an external host. When a control command is received from an external host through the host interface **1230**, the processor **1210** may control the semiconductor device **1100** in response to the control command.

(191) FIG. **24** is a schematic perspective view of a data storage system including a semiconductor device according to an example embodiment.

(192) Referring to FIG. **24**, a data storage system **2000** according to an example embodiment may include a main substrate **2001**, a controller **2002** mounted on the main substrate **2001**, one or more semiconductor packages **2003**, and a DRAM **2004**. The semiconductor package **2003** and the DRAM **2004** may be connected to the controller **2002** by interconnection patterns **2005** formed on the main substrate **2001**.

(193) The main substrate **2001** may include a connector **2006** including a plurality of pins coupled to an external host. The number and arrangement of the plurality of pins in the connector **2006** may vary according to a communications interface between the data storage system **2000** and the external host. In an implementation, the data storage system **2000** may communicate with an external host according to one of interfaces such as a universal serial bus (USB), peripheral component interconnect express (PCI-Express), serial advanced technology attachment (SATA), and M-PHY for universal flash storage (UFS). In example embodiments, the data storage system **2000** may operate by power supplied from an external host through the connector **2006**. The data storage system **2000** may further include a power management integrated circuit (PMIC) for distributing power supplied from the external host to the controller **2002** and the semiconductor package **2003**.

(194) The controller **2002** may write data in the semiconductor package **2003** or may read data from the semiconductor package **2003**, and may help improve an operation speed of the data storage system **2000**.

(195) The DRAM **2004** may be implemented by a buffer memory for reducing a difference in speed between the semiconductor package **2003**, a data storage space, and an external host. The DRAM **2004** included in the data storage system **2000** may also operate as a type of cache memory and may provide a space for temporarily storing data in a control operation performed on the semiconductor package **2003**. When the DRAM **2004** is included in the data storage system **2000**, the controller **2002** may further include a DRAM controller for controlling the DRAM **2004** in addition to the NAND controller for controlling the semiconductor package **2003**.

(196) The semiconductor package **2003** may include first and second semiconductor packages **2003***a* and **2003***b* spaced apart from each other. Each of the first and second semiconductor packages **2003***a* and **2003***b* may be configured as a semiconductor package including a plurality of semiconductor chips **2200**. Each of the semiconductor chips **2200** may include the semiconductor device described in one of the above-described example embodiments with reference to FIGS. **1** to **22**C.

(197) Each of the first and second semiconductor packages **2003***a* and **2003***b* may include a package substrate **2100**, semiconductor chips **2200** on the package substrate **2100**, adhesive layers **2300** on a lower surface of each of the semiconductor chips **2200**, a connection structure **2400** electrically connecting the semiconductor chips **2200** to the package substrate **2100**, and a molding layer **2500** covering the semiconductor chips **2200** and the connection structure **2400** on the package substrate **2100**.

(198) The package substrate **2100** may be implemented by a printed circuit board (PCB) including package upper pads **2130**. Each of the semiconductor chips **2200** may include an input/output pad **2210**.

(199) In an implementation, the connection structure **2400** may be configured as a bonding wire electrically connecting the input and output pad **2210** to the package upper pads **2130**. Accordingly,

in each of the first and second semiconductor packages **2003***a* and **2003***b*, the semiconductor chips **2200** may be electrically connected to each other by a bonding wire method and may be electrically connected to the package upper pads **2130** of the package substrate **2100**. In an implementation, in each of the first and second semiconductor packages **2003***a* and **2003***b*, the semiconductor chips **2200** may be electrically connected to each other by a connection structure including a through-silicon via (TSV), rather than the connection structure **2400** of a bonding wire method.
(200) In an implementation, the controller **2002** and the semiconductor chips **2200** may be included in a single package. In an implementation, the controller **2002** and the semiconductor chips **2200** may be mounted on an interposer substrate different from the main substrate **2001**, and the controller **2002** may be connected to the semiconductor chips **2200** through an interconnection formed on the interposer substrate.
(201) FIG. **25** is a schematic cross-sectional view of a data storage system including a semiconductor device according to an example embodiment. FIG. **25** illustrates an example embodiment of the semiconductor package **2003** of FIG. **24**, and conceptually illustrates a cross-sectional region of the semiconductor package **2003** illustrated in FIG. **24** taken along line III-III′.
(202) Referring to FIG. **25**, in the semiconductor package **2003**, the package substrate **2100** may be configured as a printed circuit board (PCB). The package substrate **2100** may include a package substrate body portion **2120**, package upper pads **2130** disposed on an upper surface of the package substrate body portion **2120**, lower pads **2125** disposed on a lower surface of the package substrate body portion **2120** or exposed through the lower surface, and internal interconnections **2135** electrically connecting the package upper pads **2130** to the lower pads **2125** in the package substrate body portion **2120**. The package upper pads **2130** may be electrically connected to the connection structures **2400**. The lower pads **2125** may be connected to interconnection patterns **2005** of a main substrate **2001** of the data storage system **2000** through conductive connection portions **2800** as illustrated in FIG. **24**.
(203) Each of the semiconductor chips **2200** may include a semiconductor substrate **3010** and a first structure **3100** and a second structure **3200** sequentially stacked on the semiconductor substrate **3010**. The first structure **3100** may include a peripheral circuit region including peripheral interconnections **3110**. The second structure **3200** may include a common source line **3205**, a gate stack structure **3210** on the common source line **3205**, memory channel structures **3220** and separation structures **3230** penetrating through the gate stack structure **3210**, bitlines **3240** electrically connected to the memory channel structures **3220**, and gate connection lines electrically connected to wordlines (WL in FIG. **23**) of the gate stack structure **3210** (**94** of FIG. **2**B). The first structure **3100** may include the first structure **1100**F of FIG. **23**, and the second structure **3200** may include the second structure **1100**S of FIG. **23**. For example, in FIG. **25**, a partially enlarged region denoted by a reference numeral **1** may represent the cross-sectional structure of FIG. **2**. Accordingly, each of the semiconductor chips **2200** may include the semiconductor device **1** according to one of the example embodiments described above with reference to FIGS. **1** to **22**C.
(204) Each of the semiconductor chips **2200** may include a through-interconnection **3245** electrically connected to the peripheral interconnections **3110** of the first structure **3100** and extending inwardly of the second structure **3200**. The through-interconnection **3245** may penetrate through the gate stack structure **3210** and may be further disposed on an external side of the gate stack structure **3210**.
(205) Each of the semiconductor chips **2200** may further include an input/output connection line **3265**, electrically connected to the peripheral interconnections **3110** of the first structure **3100** and extending inwardly of the second structure **3200**, and an input/output pad **2210** electrically connected to the input/output connection line **3265**.
(206) As described above, example embodiments may provide a method of fabricating a semiconductor device, including forming vertical memory structures and vertical dummy structures, forming an upper separation pattern, and performing an oxidation process for improving

characteristics of the vertical memory structures, a semiconductor device fabricated by the method, and a data storage system including the semiconductor device. The vertical dummy structures may be formed under the same conditions as the vertical memory structures before forming the upper separation pattern, and the vertical dummy structures may be stably formed without defects such as warpage, and the like. Thus, reliability and durability of the semiconductor device may be improved. In addition, after forming the upper separation pattern, the oxidation process for improving characteristics of the vertical memory structures may be performed to help improve data storage characteristics of the semiconductor device, e.g., performance of the semiconductor device.
(207) One or more embodiments may provide a semiconductor device exhibiting improved integration density and reliability.
(208) One or more embodiments may provide a semiconductor device exhibiting improved performance.
(209) Example embodiments have been disclosed herein, and although specific terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purpose of limitation. In some instances, as would be apparent to one of ordinary skill in the art as of the filing of the present application, features, characteristics, and/or elements described in connection with a particular embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of skill in the art that various changes in form and details may be made without departing from the spirit and scope of the present invention as set forth in the following claims.

Claims

1. A semiconductor device, comprising: a substrate structure; a stack structure on the substrate structure and including interlayer insulating layers and gate electrodes alternately and repeatedly stacked in a vertical direction that is perpendicular to an upper surface of the substrate structure; a vertical memory structure penetrating through the stack structure in the vertical direction; a vertical dummy structure penetrating through the stack structure in the vertical direction; and an upper separation pattern on the stack structure, the upper separation pattern extending in a first direction parallel to the upper s face of the substrate structure and including a first portion and a second portion, the first portion intersecting the vertical dummy structure and the second portion extending from the first portion in the first direction and penetrating through a portion of the stack structure, wherein: the second portion of the upper separation pattern penetrates through a plurality of the gate electrodes, the vertical memory structure includes an insulating region, a channel layer on a side surface of the insulating region, a first dielectric layer on an external side surface of the channel layer, a data storage layer on an external side surface of the first dielectric layer, a second dielectric layer on an external side surface of the data storage layer, and a pad pattern on the insulating region, the vertical dummy structure includes a dummy insulating region, a dummy channel layer on a side surface of the dummy insulating region, a first dummy dielectric layer on an external side surface of the dummy channel layer, a dummy data storage layer on an external side surface of the first dummy dielectric layer, a second dummy dielectric layer on an external side surface of the dummy data storage layer, and a dummy pad pattern on the dummy insulating region, and when viewed in a plan view at a first height level, higher than a height level of a lowermost end of the upper separation pattern, the dummy channel layer includes a first dummy channel region facing the dummy data storage layer and a second dummy channel region facing the dummy data storage layer, so that the dummy channel layer at the first dummy channel region has a horizontal thickness different from a horizontal thickness of the dummy channel layer at the second dummy channel region.
2. The semiconductor device as claimed in claim 1, wherein, when viewed in the plan view at the

first height level: the thickness of the dummy channel layer at the first dummy channel region is smaller than the thickness of the dummy channel layer at the second dummy channel region, and a distance between the first dummy channel region and the upper separation pattern is smaller than a distance between the second dummy channel region and the upper separation pattern.
3. The semiconductor device as claimed in claim 1, wherein, when viewed in the plan view at the first height level, the first dummy dielectric layer includes a first dummy dielectric region and a second dummy dielectric region between the dummy data storage layer and the dummy channel layer, the first dummy dielectric region having a horizontal thickness different from a horizontal thickness of the second dummy dielectric region.
4. The semiconductor device as claimed in claim 3, wherein: the horizontal thickness of the first dummy dielectric region is greater than the horizontal thickness of the second dummy dielectric region; and a distance between the first dummy dielectric region and the upper separation pattern is smaller than a distance between the second dummy dielectric region and the upper separation pattern.
5. The semiconductor device as claimed in claim 1, wherein, when viewed in the plan view at the first height level: the dummy data storage layer is divided into dummy data storage portions spaced apart from each other in a second direction by the upper separation pattern, the second direction is parallel to the upper surface of the substrate structure and is perpendicular to the first direction, and the dummy data storage layer has a substantially uniform horizontal thickness.
6. The semiconductor device as claimed in claim 1, wherein, when viewed in the plan view at the first height level, the channel layer of the vertical memory structure has a substantially uniform horizontal thickness.
7. The semiconductor device as claimed in claim 1, wherein, when viewed in the plan view at the first height level: the horizontal thickness of the dummy channel layer at the second dummy channel region is greater than the horizontal thickness of the dummy channel layer at the first dummy channel region, and a horizontal thickness of the channel layer of the vertical memory structure is greater than the horizontal thickness of the dummy channel layer at the second dummy channel region.
8. The semiconductor device as claimed in claim 1, wherein, when viewed in the plan view at the first height level: the first dummy dielectric layer includes regions having different horizontal thicknesses between the dummy data storage layer and the dummy channel layer, the first dielectric layer of the vertical memory structure has a substantially uniform horizontal thickness, and a maximum horizontal thickness of the first dummy dielectric layer is greater than a horizontal thickness of the first dielectric layer of the vertical memory structure.
9. The semiconductor device as claimed in claim 1, wherein, when viewed in the plan view at the first height level: the dummy channel layer includes a first dummy channel portion and a second dummy channel portion spaced apart from each other in a second direction, the second direction is parallel to the upper surface of the substrate structure and is perpendicular to the first direction, and at least one of the first dummy channel portion and the second dummy channel portion includes the first dummy channel region and the second dummy channel region.
10. The semiconductor device as claimed in claim 9, wherein, when viewed in the plan view at a second height level, the second height level being higher than the height level of the lowermost end of the upper separation pattern and lower than the first height level: the dummy channel layer has a ring shape, the dummy channel layer includes a third dummy channel region disposed in the first direction from a center of the vertical dummy structure and a fourth dummy channel region disposed in a second direction from the center of the vertical dummy structure, the second direction is perpendicular to the first direction, and a horizontal thickness of the dummy channel layer at the fourth dummy channel region is greater than a horizontal thickness of the dummy channel layer at the third dummy channel region.
11. The semiconductor device as claimed in claim 1, wherein at least a portion of the dummy pad

pattern is divided into pad portions spaced apart from each other in a second direction, perpendicular to the first direction, by the first portion of the upper separation pattern.

12. The semiconductor device as claimed in claim 1, wherein: the upper separation pattern further includes an upper portion covering an upper surface of the stack structure, and the first portion, the second portion, and the upper portion are integrated with each other.

13. The semiconductor device as claimed in claim 1, further comprising: a first oxide layer; a second oxide layer; separation structures; a bitline contact plug; a bitline; and a gate dielectric layer covering an upper surface and a lower surface of each of the gate electrodes and interposed between the vertical memory structure and the gate electrodes, between the vertical dummy structure and the gate electrodes, and between the upper separation pattern and the gate electrodes, wherein: the substrate structure is a semiconductor substrate, the vertical memory structure further includes a channel pattern in contact with the semiconductor substrate, extending upwardly, and in contact with the channel layer, the vertical dummy structure further includes a dummy channel pattern in contact with the semiconductor substrate, extending upwardly, and in contact with the dummy channel layer, upper surfaces of the channel pattern and the dummy channel pattern are at a first height level higher than a height level of a lowermost gate electrode, the first oxide layer is in contact with the channel pattern and between the lowermost gate electrode and the channel pattern, the second oxide layer is in contact with the dummy channel pattern and between the lowermost gate electrode and the dummy channel pattern, the separation structures penetrate through the stack structure and extend in the first direction, the upper separation pattern, the vertical memory structure, and the vertical dummy structure are between the separation structures, the bitline contact plug is in contact with the pad pattern of the vertical memory structure, and the bitline is electrically connected to the bitline contact plug.

14. The semiconductor device as claimed in claim 1, wherein: the substrate structure includes a semiconductor substrate, a peripheral circuit region on the semiconductor substrate, and an upper substrate on the peripheral circuit region; the upper substrate includes at least one silicon layer; and the vertical memory structure and the vertical dummy structure are in contact with the at least one silicon layer.

15. The semiconductor device as claimed in claim 1, further comprising an upper semiconductor chip on the stack structure, wherein: the upper semiconductor chip includes a semiconductor substrate and a peripheral circuit region below the semiconductor substrate; and the peripheral circuit region is between the semiconductor substrate and the stack structure.

16. A semiconductor device, comprising: a substrate structure; a stack structure on the substrate structure and including interlayer insulating layers and gate electrodes alternately and repeatedly stacked in a vertical direction that is perpendicular to an upper surface of the substrate structure; a vertical memory structure penetrating through the stack structure in the vertical direction; a vertical dummy structure penetrating through the stack structure in the vertical direction; an upper separation pattern on the stack structure, the upper separation pattern including a first portion and a second portion extending in a first direction, parallel to the upper surface of the substrate structure, the first portion intersecting the vertical dummy structure and the second portion extending from the first portion and penetrating through a portion of the stack structure; a contact plug in contact with the vertical memory structure and on the vertical memory structure; and a bitline electrically connected to the contact plug and on the contact plug, wherein: the second portion of the upper separation pattern penetrates through each of a plurality of upper gate electrodes of the gate electrodes, the vertical memory structure includes an insulating region, a channel layer contacting a side surface of the insulating region, a first dielectric layer on-contacting an external side surface of the channel layer, a data storage layer on an external side surface of the first dielectric layer, a second dielectric layer on an external side surface of the data storage layer, and a pad pattern on the insulating region, the vertical dummy structure includes a dummy insulating region, a dummy channel layer contacting a side surface of the dummy insulating region, a first dummy dielectric

layer contacting an external side surface of the dummy channel layer, a dummy data storage layer on an external side surface of the first dummy dielectric layer, a second dummy dielectric layer on an external side surface of the dummy data storage layer, and a dummy pad pattern on the dummy insulating region, and when viewed in a plan view at a first height level higher than a height level of a lowermost surface of a lowermost one of the plurality of upper gate electrodes, and lower than a height level of a lowermost surface of the pad pattern, a horizontal thickness of the dummy channel layer of the vertical dummy structure is smaller than a horizontal thickness of the channel layer of the vertical memory structure.

17. The semiconductor device as claimed in claim 16, wherein, when viewed in the plan view at the first height level: the dummy data storage layer includes dummy data storage portions spaced apart from each other in a second direction, perpendicular to the first direction, the dummy channel layer includes dummy channel portions spaced apart from each other in the second direction, and at least one of the dummy channel portions includes a dummy channel region having a gradually increased width.

18. The semiconductor device as claimed in claim 16, wherein, when viewed e plan view at the first height level: the dummy channel layer of the vertical dummy structure includes a first dummy channel region having a first minimum horizontal thickness and a second dummy channel region having a first maximum horizontal thickness, the channel layer has a substantially uniform horizontal thickness, and the first maximum horizontal thickness of the second dummy channel region is smaller than the horizontal thickness of the channel layer of the vertical memory structure.

19. The semiconductor device as claimed in claim 18, wherein, at a second height level that is higher than a height level of a lowermost surface of a lowermost one of the upper gate electrodes, and that is higher than the first height level: the dummy channel layer has a ring shape, the dummy channel layer includes a third dummy channel region having a second minimum horizontal thickness and a fourth dummy channel region having a second maximum horizontal thickness, and the second minimum horizontal thickness of the third dummy channel region is greater than the first minimum horizontal thickness of the first dummy channel region.

20. A data storage system, comprising: a semiconductor device including an input/output pad; and a controller electrically connected to the semiconductor device through the input/output pad and configured to control the semiconductor device, wherein: the semiconductor device includes: a substrate structure, a stack structure on the substrate structure and including interlayer insulating layers and gate electrodes alternately and repeatedly stacked in a vertical direction, perpendicular to an upper surface of the substrate structure, a vertical memory structure penetrating through the stack structure in the vertical direction, a vertical dummy structure penetrating through the stack structure in the vertical direction, an upper separation pattern on the stack structure, the upper separation pattern extending in a first direction parallel to the sort of the substrate structure and including a first portion and a second portion, the first portion intersecting the vertical dummy structure and the second portion extending in the first direction from the first portion and penetrating through a portion of the stack structure, a contact plug in contact with the vertical memory structure and on the vertical memory structure, and a bitline electrically connected to the contact plug and on the contact plug, the second portion of the upper separation pattern penetrates through each of a plurality of upper gate electrodes of the gate electrodes, the vertical memory structure includes an insulating region, a channel layer on a side surface of the insulating region, a first dielectric layer on an external side surface of the channel layer, a data storage layer on an external side surface of the first dielectric layer, a second dielectric layer on an external side surface of the data storage layer, and a pad pattern on the insulating region, the vertical dummy structure includes a dummy insulating region, a dummy channel layer on a side surface of the dummy insulating region, a first dummy dielectric layer on an external side surface of the dummy channel layer, a dummy data storage layer on an external side surface of the first dummy dielectric layer, a second dummy dielectric layer on an external side surface of the dummy data storage layer, and a

dummy pad pattern on the dummy insulating region, when viewed in a plan view at a first height level that is higher than a height level of a lowermost surface of a lowermost one of the plurality of upper gate electrodes, and lower than a height level of a lowermost surface of the pad pattern, the dummy channel layer of the vertical dummy structure includes a first dummy channel region having a first minimum horizontal thickness and a second dummy channel region having a first maximum horizontal thickness, when viewed in the plan view at the first height level, the channel layer has a substantially uniform horizontal thickness, and the first maximum horizontal thickness of the dummy channel layer at the second dummy channel region is smaller than the horizontal thickness of the channel layer of the vertical memory structure.